

Zhuoyuan Li

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Education

Purdue University, West Lafayette

B.S. in Computer Science, expected May 2027 · August 2025–present

GPA: 3.88/4.00 (through Summer 2026). Dean's List and Semester Honors, Fall 2025 and Spring 2026.

Rensselaer Polytechnic Institute

Computer Science · August 2023–May 2025

GPA: 3.77/4.00. Dean's Honor List: Spring 2024, Fall 2024, Spring 2025.

Publication

LEO Satellite Network Resilience Analysis: A Focus on Critical Satellites

Zhuoyuan Li*, Wenyi (Morty) Zhang*, Wenhao Chen, Yiyang Hu, Weyi Lu. * Equal contribution.

ACM LEO-NET 2024, co-located with ACM MobiCom. doi:10.1145/3697253.3697267

- Extended Hypatia-based simulations to identify heavily used satellites and evaluate targeted failures in Starlink and Kuiper shells.
- Analyzed satellite usage redistribution and RTT changes after removing selected satellites and recomputing city-pair paths.

Research

Testing the Transferability of Critical-Satellite Rankings in LEO Networks

Zhuoyuan Li · Manuscript, 2026 · PDF

- Built a paired-deletion pipeline to compare damage-based rankings across routing policies, using a common satellite universe and matched traffic flows.
- Evaluated transfer across 24 orbital phases and multiple ECMP hashes; identified a confirmed peak-link-stress transfer loss from utilization-aware routing to IGP-ECMP.
- Tested traffic-based prediction and topology expansion as explanations or mitigation; retained their negative results and narrowed the claim accordingly.

CLIP Adaptation for Disaster-Response Classification · Purdue VIP AI4SMA, Spring 2026

- Analyzed cross-modal label inconsistency in CrisisMMD; implemented agreement-based filtering and consolidated eight humanitarian categories into five.
- Compared head-only training, LoRA, and full fine-tuning of CLIP ViT-B/32 on agreed-only data splits.
- Head-only training achieved 93.2% accuracy / 92.2% macro-F1 on binary Task 1 and 89.3% / 83.9% on the consolidated five-class Task 2.

Selected Engineering Work

Credit Card Fraud Detection Pipeline [Code]

- Built XGBoost training with chronological data splits, FastAPI batch scoring, and request/latency monitoring; implemented PSI drift checks for transaction amounts and model scores.

Technical Skills

Python, C++, C, Java, R, MATLAB · NumPy, Pandas, scikit-learn, XGBoost, FastAPI · Git, Linux, Conda

Network simulation, graph-based failure analysis, multimodal classification, model adaptation, evaluation under class imbalance